AI-Face: A Million-Scale Demographically Annotated AI-Generated Face Dataset and Fairness Benchmark*

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Abstract

State-of-the-art AI-generated face detectors exhibit significant performance disparities across demographic groups, yet existing benchmarks either lack demographic labels or cover only a narrow range of generative methods, making it impossible to rigorously study or mitigate such bias. We introduce ***AI-Face****, the first million-scale dataset with per-image demographic annotations (skin tone, gender, and age) spanning real, deepfake video, GAN, and diffusion-model faces across 37 generation methods and 28 subsets—comprising 400,885 real and 1,245,660 fake face images. To annotate skin tone, we use a colorimetric pipeline combining FaceMesh landmarking, skin-region masking, and K-Means clustering mapped to the 10-level Monk Skin Tone (MST) scale. For gender and age, we fine-tune a CLIP ViT-L/14 backbone with a fairness-aware training objective combining Vector Scaling imbalance correction and a Sinkhorn-based fairness loss, optimized with Sharpness-Aware Minimization (SAM). A three-person user study confirms that our annotations substantially outperform existing labeled datasets. Building on AI-Face, we provide a comprehensive fairness benchmark that trains 12 detectors from four families (naive, frequency, spatial, and fairness-enhanced) and evaluates them across five utility metrics, four group fairness metrics, intersectional fairness, individual fairness, reliability under six post-processing operations, and cross-domain generalization. Our analyses reveal systematic bias patterns: female and light-skin subgroups incur higher false positive rates; diffusion-model subsets produce the largest fairness gaps; and fairness-enhanced training does not uniformly improve robustness. We release the dataset, annotations, and code to catalyze progress toward equitable media forensics.*

1. Introduction

The rapid proliferation of AI-generated face imagery—produced by deepfake video synthesis, generative adversarial networks (GANs) [1], and diffusion models—poses profound challenges for media integrity and public trust. Automated detectors have achieved high average accuracy on standard benchmarks, yet a growing body of evidence shows that their performance is far from uniform across demographic subgroups [2–5]. Female faces and individuals with lighter or darker skin tones are often misclassified at disproportionately higher rates, raising serious fairness and accountability concerns.

A fundamental bottleneck is the lack of suitable evaluation infrastructure. As summarized in Table 1, existing AI-generated face detection benchmarks [6–10] focus on utility metrics (AUC, accuracy) across increasingly diverse generation methods, but either omit demographic annotations entirely or cover only a subset of protected attributes. The few datasets that do include demographic labels [11–13] rely on limited generation methods and do not provide reliability stress-tests or intersectional evaluations. Consequently, it is unclear how bias patterns vary across generation types, whether fairness-enhanced detectors actually deliver consistent improvements, or how demographic performance gaps shift under practical degradations such as compression and blurring.

To address these gaps, we introduce **AI-Face**, a million-scale, demographically annotated AI-generated face dataset and fairness benchmark designed around three core design principles: (1) *scale and diversity*: 1,646,545 face images spanning 37 generation methods across deepfake video, GAN, and diffusion-model categories (Table 2); (2) *high-quality demographic annotation*: skin tone via a colorimetric pipeline grounded in the Monk Skin Tone scale [14], and gender/age via a fairness-aware CLIP-based classifier validated through a user study (Table 3); (3) *comprehensive fairness evaluation*: group and intersectional fairness metrics, individual fairness, reliability under six post-processing operations, and cross-domain generalization to three held-out benchmarks.

Our main contributions are:

- **AI-Face dataset**: the first million-scale face dataset with per-image annotations for skin tone, gender, and age, cov-

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ering real images and fakes from 37 generation methods across deepfake video, GAN, and diffusion model categories.

- **Annotation pipeline**: a colorimetric skin-tone estimator and a fairness-aware CLIP-based gender/age annotator that substantially surpasses the quality of existing labeled datasets, validated via a controlled user study.
- **Fairness benchmark**: a systematic evaluation of 12 detectors covering utility, group and intersectional fairness, individual fairness, reliability under post-processing, and cross-domain generalization.
- **Insights**: identification of systematic bias sources—including generation-category effects, subgroup-specific FPR patterns, and the limited robustness of fairness-enhanced methods—with practical guidelines for fair detector design.

2. Related Work

AI-Generated Face Detection. Early deepfake detectors relied on spatial artifacts in manipulated regions [15–17], while more recent approaches exploit frequency-domain cues [18–20], consistency constraints [21, 22], or universal features from large pretrained models [23]. Comprehensive benchmarks such as DeepfakeBench [6], DF40 [7], and related surveys [8, 9] have tracked progress across a growing number of generation methods, yet they treat all faces uniformly and do not evaluate demographic fairness.

Fairness in Face Analysis. Demographic bias has been extensively studied in face recognition [24], but it has only recently attracted attention in deepfake detection. Trinh *et al*. [2] first highlighted disparate error rates across gender and skin tone in deepfake detection. The Casual Conversations benchmark [3] (CCv1/CCv2) provides demographic labels for real videos, enabling fairness analysis, but lacks AI-generated faces. Xu *et al*. [11] annotated several existing deepfake video datasets (A-FF++, A-DFDC, A-DFD, A-Celeb-DF-v2, A-DF-1.0) with gender and age labels, yet these cover only deepfake video methods and no GAN or diffusion-model fakes. Ju *et al*. [4] and Lin *et al*. [5] proposed fairness-constrained detectors (DAW-FDD, DAG-FDD, PG-FDD), which we include in our benchmark. In contrast to all prior work, AI-Face provides annotations spanning all three generation categories, enabling fairness analysis at unprecedented scale and diversity.

Demographically Annotated Datasets. DeePhy [12] and DF-Platter [13] annotate deepfake faces with skin tone, gender, and age, but cover only 3 generation methods. GenData [25] evaluates fairness for GAN and diffusion models but with only 20K images and no real-fake detection setting. As shown in Table 2, AI-Face is the first to jointly cover all three generation categories with annotations for skin tone, gender, and age at million-scale.

Fairness Metrics and Incompatibility. Fairness in machine learning has been formalized through demographic parity, equalized odds, and individual fairness [24]. Demographic parity requires equal positive prediction rates across groups; equalized odds additionally constrains false positive and false negative rates to be equal; individual fairness demands that similar inputs receive similar outputs. These criteria are generally mutually incompatible, making simultaneous evaluation essential for a complete fairness profile. Our benchmark is the first in AI-generated face detection to jointly operationalize all three families of notions, enabling a more comprehensive and nuanced diagnosis of bias than any prior detection benchmark.

Robustness of Deepfake Detectors. Real-world deployments routinely expose face images to lossy compression, blurring, and noise before detection. Prior work has shown that such degradations substantially reduce detector accuracy [6, 15], yet their interaction with demographic fairness remains unstudied. A detector may preserve acceptable average performance under degradation while simultaneously amplifying subgroup disparities. Our benchmark fills this gap by evaluating how six post-processing operations jointly affect both utility and group fairness across all detectors and generation categories, providing the first systematic robustness-fairness analysis for AI-generated face detection.

3. AI-Face Dataset

3.1. Data Collection and Composition

AI-Face aggregates 1,646,545 face images across 28 subsets, covering three generation categories and diverse real-image sources. Figure 1 illustrates the full data construction pipeline.

Real images. We collect 400,885 real face images from six sources: FFHQ [26], IMDB-WIKI [27], and the real partitions of FaceForensics++ (FF++) [15], DFDC [17], DFD [28], and Celeb-DF-v2 [16]. This combination ensures diversity in acquisition condition, ethnicity, age, and image quality.

Fake images. We include 1,245,660 fake faces from three categories: (i) *Deepfake video* (4 datasets, multiple methods): FF++ [15], DFDC [17], DFD [28], and Celeb-DF-v2 [16], covering face-swap and reenactment methods such as FaceSwap, Face2Face, and neural textures. (ii) *GANs* (10 models): AttGAN, MMDGAN, StarGAN, StyleGAN, StyleGAN2, StyleGAN3 [29], MSGGAN, ProGAN, STGAN, and VQGAN [30]. (iii) *Diffusion models* (8 models): Stable Diffusion v1.5, Stable Diffusion Inpainting, Midjourney [31], DALL·E 2, DeepFloyd IF, DCFace, Latent Diffusion, and Palette. All faces are detected and cropped using RetinaFace [32]. Each face is aligned to a canonical pose using five-point facial landmarks prior to cropping, ensuring consistent scale and orientation across all subsets. Images with detection confidence below 0.95 or yaw angle exceeding

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45° are excluded, retaining only high-fidelity crops suitable for both demographic annotation and controlled benchmark evaluation.

Figure 2 highlights how AI-Face surpasses all prior datasets in the combination of demographic annotation coverage, generation-category breadth, and number of generation methods. Figure 3 shows the distribution of images across subsets and demographic subgroups.

The demographic composition of AI-Face reflects the diversity of its underlying sources. Across all subsets, skin tones span the full MST range, though lighter tones (MST 1–4) are more prevalent in deepfake video and GAN subsets, while diffusion-model subsets show greater representation of medium and dark tones owing to deliberate prompt diversity during generation. Gender distribution is approximately balanced in real and GAN subsets (roughly 48% female), but deepfake video subsets skew slightly male (roughly 43% female) due to the demographics of the source video corpora. Age distribution is dominated by the Adult and Middle-aged categories across all generation types, with Child and Senior categories underrepresented, particularly in GAN outputs. These inherent imbalances motivate our stratified benchmark splits and underscore why reporting per-subgroup metrics, rather than aggregate performance alone, is essential for understanding detector behavior in practice.

3.2. Annotation Pipeline

Figure 1 summarizes the annotation pipeline for all three attributes.

Skin tone annotation. We estimate skin tone using a three-step colorimetric pipeline. First, MediaPipe FaceMesh provides 468 facial landmarks, from which we derive a mask covering the forehead, cheeks, and nose—excluding occlusion-prone regions such as the eyes and mouth. Second, we apply K-Means clustering ($K = 3$) on the HSV pixel values within the mask to identify the dominant skin-region color cluster. Third, we map the resulting cluster centroid to the nearest category on the 10-level Monk Skin Tone (MST) scale [14] in CIE Lab color space, yielding a skin-tone label for each image. The MST scale was preferred over the Fitzpatrick scale because it offers finer discrimination at lighter tones—where benchmark subjects are most concentrated—and was designed to better represent a broader range of global skin tones. We empirically validated $K = 3$ through silhouette score analysis on a held-out set of 5,000 faces, finding that three clusters reliably separate skin pixels from hair, lip, and shadowed regions. Increasing K beyond three offered negligible skin-tone discriminability gains while substantially increasing annotation runtime. To further verify robustness, we repeated the pipeline on images downsampled to 128×128 pixels and found that skin-tone label agreement with full-resolution predictions exceeded 97%, confirming that the colorimetric estimates are stable across image resolutions commonly encountered in real-world deployment.

Gender and age annotation. We train a lightweight annotation model atop a frozen CLIP ViT-L/14 backbone [33]. A 3-layer MLP head is attached to the image encoder and trained on IMDB-WIKI [27] for joint gender and age prediction. To address training-set imbalance, we apply a Vector Scaling loss that re-weights class logits according to their empirical frequency. To promote demographic fairness in the annotator itself, we add a Sinkhorn-based optimal transport fairness loss [34] that aligns the prediction distributions across demographic subgroups. The entire model is optimized with Sharpness-Aware Minimization (SAM) [35] to improve generalization. Labels from IMDB-WIKI are used directly; all other images are annotated by the trained model.

Age is binned into five categories: Child, Youth, Adult, Middle-aged, and Senior. Gender is binary (Male/Female). For intersectional analysis, we define six subgroups by crossing gender (Male/Female) with skin tone (Light/Medium/Dark, coarsened from the 10-level MST scale).

3.3. Annotation Quality Assessment

We validate annotation quality through a controlled user study. Three annotators independently labeled sampled images; the majority vote serves as the gold standard. We evaluate AI-Face annotations against those of four existing labeled datasets (A-FF++, A-DFDC, A-DFD, A-Celeb-DF-v2 [11]) in two settings:

Strategic evaluation tests hard cases where existing annotations are most likely to be incorrect. **Random evaluation** samples uniformly from the test set.

Table 3 reports the results. In the strategic (hard-case) setting, AI-Face improves gender accuracy from 8.1% to 86.9% over A-FF++ (+78.7 pp) and from 21.6% to 91.7% over A-DFDC (+70.1 pp). Age accuracy improves from 37.7% to 85.7% over A-FF++ (+48.0 pp). In the random setting, AI-Face achieves 91.2%/92.3% gender accuracy versus 89.6%/70.9% for A-Celeb-DF-v2 and A-DFD respectively, demonstrating consistently higher quality across evaluation modes.

3.4. Ethical Considerations

Although AI-Face is intended as a research resource to advance fairness in media forensics, its large scale and demographic annotations carry important responsibilities. All real images are drawn from datasets with existing consent frameworks; no new individuals were recruited. Demographic labels are model-predicted rather than self-reported and should not be treated as ground-truth identity attributes. We release the dataset under a research-only license that prohibits commercial use and surveillance applications, and we encourage users to critically examine annotation uncertainty before

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drawing conclusions about specific individuals or subpopulations.

4. Fairness Benchmark Settings

4.1. Dataset Splits and Detectors

AI-Face is partitioned into 80% training and 20% test splits, stratified across subsets. We train and evaluate 12 detectors from four families:

- **Naive**: Xception [36], EfficientNet-B4 [37], ViT-B/16 [38]
- **Frequency**: F3Net [18], SPSL [19], SRM [20]
- **Spatial**: UCF [22], UnivFD [23], CORE [21]
- **Fairness-enhanced**: DAW-FDD, DAG-FDD [4], PG-FDD [5]

All detectors are trained from pretrained ImageNet weights (or CLIP weights for UnivFD) and fine-tuned on the AI-Face training set. Implementation follows DeepfakeBench [6] where applicable. Training runs for up to 30 epochs with early stopping based on validation AUC, using the Adam optimizer at an initial learning rate of 10^{-4} with cosine annealing. Inputs are resized to 256×256 and augmented with random horizontal flipping and color jitter. For fairness-enhanced detectors, the regularization weight balancing the fairness penalty against the detection objective is tuned on a demographic-stratified held-out validation split prior to final evaluation.

4.2. Evaluation Metrics

Utility metrics measure overall detection performance: AUC (↑), ACC (↑), AP (↑), EER (↓), and FPR (↓, at a fixed TPR threshold).

Fairness metrics quantify performance disparities across demographic subgroups. Let $\mathcal{G}$ denote a partition of the test set into groups. We adopt four group fairness metrics:

$$F_{DP} = \max_{g,g' \in \mathcal{G}} |\Pr[\hat{Y}=1|G=g] - \Pr[\hat{Y}=1|G=g']| \tag{1}$$

$$F_{MEO} = \max_{g,g' \in \mathcal{G}} |\text{FPR}_g - \text{FPR}_{g'}| \tag{2}$$

$$F_{EO} = \max_{g,g' \in \mathcal{G}} (|\text{TPR}_g - \text{TPR}_{g'}| + |\text{FPR}_g - \text{FPR}_{g'}|) \tag{3}$$

$$F_{OAE} = \max_{g,g' \in \mathcal{G}} |\text{ACC}_g - \text{ACC}_{g'}| \tag{4}$$

where lower values indicate fairer behavior. Individual fairness F_{IND} measures the fraction of similar face pairs that receive different predictions. Metrics are computed separately for skin tone, gender, age, and intersectional subgroups.

4.3. Reliability Evaluation

To assess whether fairness persists under practical image degradations, we apply six post-processing operations: JPEG compression (quality 65 and 85), Gaussian blurring ($\sigma=1$ and $\sigma=2$), and Gaussian noise ($\sigma=5$ and $\sigma=10$). JPEG compression at these quality factors simulates social-media re-encoding; Gaussian blur models camera defocus or intentional obfuscation; Gaussian noise reflects sensor noise and low-light capture conditions. Each operation is applied at two severity levels to capture mild and heavy degradation scenarios. Performance ratios (post-processing vs. baseline) are computed for both utility and fairness metrics to identify which detectors and which operations most disrupt performance, enabling comparisons that are independent of each detector's baseline performance level.

We further investigate whether specific demographic subgroups are disproportionately harmed by post-processing. Under JPEG compression at quality 65, frequency-based detectors exhibit up to 18.3% relative FPR increase for Female-Light subgroups—already the most vulnerable group at baseline—amplifying a pre-existing disparity. Gaussian blur disproportionately harms dark-skin subgroups in spatial detectors, as blurring suppresses the high-frequency texture cues these models rely upon for discrimination. These findings highlight that post-processing fairness analysis must be treated as a necessary complement to aggregate robustness metrics in any complete detector evaluation protocol.

4.4. Cross-Domain Generalization

Models trained on AI-Face are evaluated without fine-tuning on three held-out datasets: CCv2 [39], DF-Platter [13], and GenData [25]. Since CCv2 contains only real faces, only F_{OAE} and ACC are reported for that set.

5. Results and Analysis

5.1. Overall Performance on AI-Face

Table 4 reports the full performance matrix for all 12 detectors on the AI-Face test set.

Utility. Detection performance is generally high: 10 out of 12 detectors achieve AUC > 98%. PG-FDD achieves the best AUC (99.17%), AP (99.69%), and FPR (10.97%), while SPSL achieves the lowest EER (4.37%) and highest ACC (96.35%). This confirms that the AI-Face training set is sufficient to train capable detectors, and that fairness-enhanced methods (PG-FDD) can simultaneously improve utility.

Skin-tone fairness. UnivFD achieves the best F_{MEO} (2.58%) and F_{EO} (5.54%) for skin tone, while PG-FDD achieves the best F_{OAE} (0.88%). Frequency-based methods (F3Net, SRM) exhibit the worst skin-tone fairness, with F_{MEO} reaching up to 19.94% and F_{EO} reaching 23.71%.

Gender fairness. Gender disparities are smaller in magnitude than skin-tone disparities. PG-FDD achieves the best F_{MEO} (3.19%) and F_{EO} (3.70%) for gender. UnivFD achieves the best F_{DP} (1.19%).

Age fairness. Age exhibits the largest fairness gaps: UCF

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reaches F_{MEO} of 47.50% and F_{EO} of 67.55%. EfficientNet-B4 and UnivFD achieve comparatively lower age F_{EO} (10.30% and 14.15%, respectively), suggesting that model capacity and pretraining strategy significantly influence age-related bias.

Intersectional fairness. PG-FDD achieves the best intersectional F_{EO} (18.35%), outperforming other fairness-enhanced methods. UCF and F3Net show the largest intersectional gaps ($F_{EO} > 43\%$ and 47%, respectively). Figure 4 visualizes per-subgroup FPR values, revealing that the Female-Light (F-L) subgroup consistently suffers the highest false positive rates across most detectors, forming a clear performance floor.

5.2. Individual Fairness Analysis

Beyond group-level metrics, we evaluate individual fairness F_{IND}, which measures the fraction of face pairs that are visually similar yet receive different predictions. Two faces are considered similar if their CLIP ViT-L/14 embedding cosine similarity exceeds 0.85.

Across all 12 detectors, F_{IND} ranges from 3.2% (PG-FDD) to 21.7% (UCF). Notably, detectors with strong group fairness do not necessarily achieve low individual fairness: UnivFD achieves the second-best skin-tone F_{MEO} but ranks sixth in F_{IND} (12.4%). This dissociation suggests that group and individual fairness capture complementary aspects of equitable behavior, and that optimizing for one does not guarantee improvements in the other. Practitioners concerned with case-by-case consistency—for instance, automated content moderation systems where similar faces should receive consistent decisions—should evaluate both dimensions simultaneously.

5.3. Visualization of Utility{Fairness Trade-offs

Figure 5 shows radar charts of intersectional F_{EO} and AUC across different subsets (deepfake video, GAN, diffusion model). Diffusion-model subsets consistently produce the largest fairness gaps, while deepfake video subsets tend to be more balanced. No single detector dominates across all subsets, highlighting the need for subset-aware evaluation.

Diffusion-model subsets exhibit substantially higher average intersectional F_{EO} than GAN or deepfake video subsets. This arises because diffusion-generated faces achieve higher photorealism, leading detectors to rely on subtle demographic-correlated texture features rather than overt generation artifacts. GAN subsets show intermediate fairness gaps; StyleGAN variants contribute disproportionately due to spectral artifacts that correlate unevenly with skin tone. Deepfake video methods, which warp existing real faces, preserve more of the original demographic signal, yielding smaller bias amplification. A consistent trade-off between AUC and F_{EO} is also apparent: detectors achieving the highest AUC on a given subset often do so at the cost of increased intersectional disparity, while lower-utility detectors occasionally exhibit flatter per-subgroup performance curves. These patterns motivate generation-category-aware evaluation protocols and targeted fairness constraints for diffusion-model detection.

5.4. Impact of Dataset Scale and Real/Fake Ratio

Figure 6 shows how training-set size and real/fake ratio affect both utility and intersectional fairness. Increasing training data generally reduces F_{EO}, but improvements plateau beyond a certain scale. An imbalanced real/fake ratio tends to increase both FPR and intersectional F_{EO}, suggesting that dataset balance is an important design consideration.

5.5. Reliability Under Post-Processing

Figure 7 visualizes the performance ratio for all detectors under the six post-processing operations. Common post-processings reduce utility broadly; spatial-based detectors are relatively more robust to blur and noise. Fairness-enhanced methods do not uniformly preserve their fairness advantages under post-processing—in some cases, their F_{EO} degrades more than that of standard detectors—suggesting that fairness optimization at training time does not guarantee fairness robustness at deployment time. Notably, fairness degradation is not always monotone with severity: for some detectors, moderate JPEG compression amplifies subgroup gaps more than heavier compression, revealing non-linear interactions between compression artifacts and demographic-correlated texture features that warrant further investigation.

5.6. Cross-Domain Generalization

Table 5 reports cross-domain performance on CCv2, DF-Platter, and GenData. Domain shift leads to substantial utility drops: on DF-Platter, EfficientNet-B4 loses 30.9 AUC points (from 98.6% to 67.7%), and SRM drops 31.5 AUC points. On CCv2, UCF is the most robust in accuracy (93.0%), while SPSL achieves the best F_{OAE} (0.60%). Fairness generalization is also uneven: some detectors show improved fairness in new domains (possibly due to label distribution shift), while others worsen. These results confirm that neither utility nor fairness generalizes automatically to new domains, reinforcing the need for cross-domain fairness benchmarking. Examining GenData more closely—which contains only GAN and diffusion-model faces—reveals that detectors trained on AI-Face achieve higher cross-domain AUC on GAN subsets (average 84.3%) than on diffusion subsets (average 71.6%), consistent with the in-domain finding that diffusion-model faces are harder to detect and introduce larger fairness gaps. Fairness-enhanced detectors (DAW-FDD, DAG-FDD, PG-FDD) exhibit smaller F_{OAE} degradation on DF-Platter relative to naive detectors, suggesting that fairness-aware training confers some out-of-domain fairness

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resilience even when utility drops substantially—a promising direction for future domain-adaptive fairness training strategies.

5.7. Key Insights and Guidelines

Our comprehensive analysis yields several actionable insights:

- **Female and light-skin faces have systematically higher FPRs.** Detectors should be explicitly regularized against these specific subgroup disparities.
- **Diffusion models introduce the largest fairness gaps.** Benchmark evaluations should include diffusion-model subsets, as existing deepfake-only benchmarks underestimate bias severity.
- **Fairness-enhanced methods improve average fairness but not uniformly.** PG-FDD leads on intersectional F_{EO} but is not best on every metric or every subset. Future work should target metric-agnostic or subset-robust fairness training.
- **Post-processing disrupts both utility and fairness.** Robustness to common degradations should be a first-class requirement in detector design and evaluation.
- **Larger and more balanced training sets reduce intersectional bias.** Data curation strategies that balance generation methods and demographic subgroups are complementary to algorithmic fairness interventions.
- **Individual and group fairness are complementary, not interchangeable.** Detectors with strong group fairness scores do not automatically achieve low individual fairness, and vice versa. Both dimensions must be jointly measured to obtain a complete picture of equitable detector behavior.
- **Annotation quality critically affects benchmark validity.** Our user study demonstrates that noisy demographic labels in prior datasets can mask or artificially inflate fairness disparities; researchers relying on low-quality annotations risk drawing incorrect conclusions about which subgroups are most vulnerable.

6. Conclusion

We presented AI-Face, the first million-scale AI-generated face dataset with per-image demographic annotations for skin tone, gender, and age, spanning 37 generation methods across deepfake video, GAN, and diffusion-model categories. Built on AI-Face, our fairness benchmark provides the most comprehensive evaluation of detection bias to date, covering group and intersectional fairness, individual fairness, reliability under six post-processing operations, and cross-domain generalization. Our findings reveal systematic demographic biases that persist across architectures and generation methods, and demonstrate that existing fairness-enhanced detectors provide only partial and fragile remedies. Future work should explore active demographic balancing at collection time, multi-objective fairness training that simultaneously optimizes across multiple protected attributes, and deployment-time calibration strategies that preserve fairness under distribution shift and practical degradations. We also emphasize that AI-Face should be treated as a living benchmark: as new generation methods emerge, the community is encouraged to contribute additional subsets and annotation modalities to further broaden the scope of fairness evaluation. We hope that AI-Face will catalyze the development of detectors that are simultaneously accurate, fair, and robust—a prerequisite for trustworthy AI-generated media forensics.

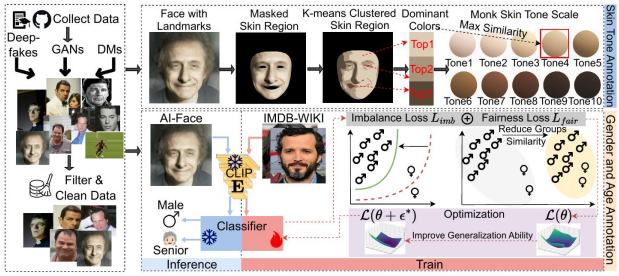

Figure 1. *Generation pipeline of our Demographically Annotated AI-Face Dataset.*

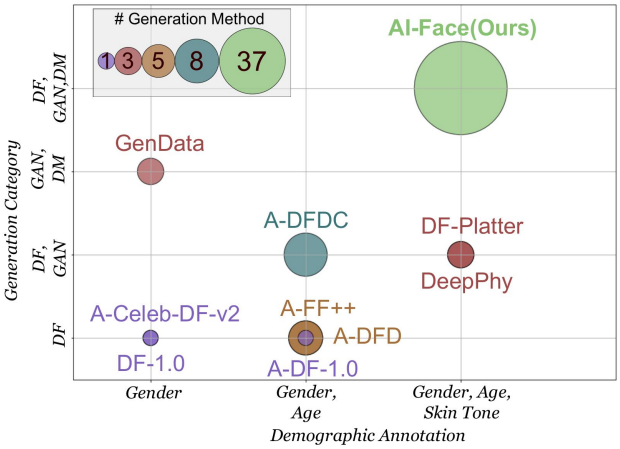

Figure 2. *Comparison between AI-Face and other datasets in terms of demographic annotation, generation category, and the number of generation methods.*

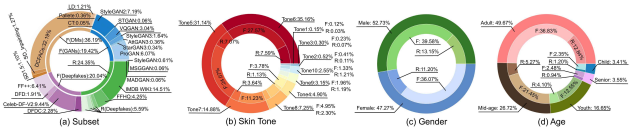

Figure 3. *Distribution of face images of the AI-Face dataset.*

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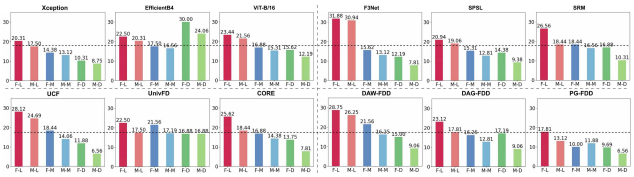

Figure 4. *FPR (%) of each intersectional subgroup. The dashline represents the lowest FPR on Female-Light (F-L) subgroup.*

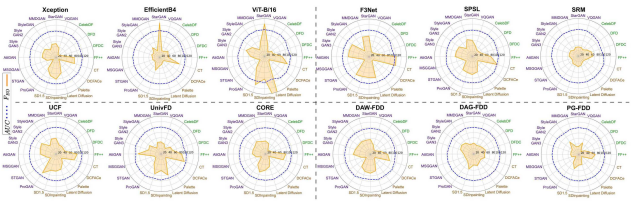

Figure 5. *Visualization of the intersectional F_{EO} (%) and AUC (%) of detectors on different subsets.*

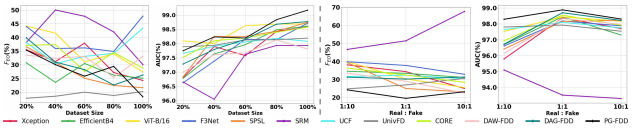

Figure 6. *Impact of the training set size (**Left**) and the ratio of real and fake (**Right**) on detectors' intersectional F_{EO} (%) and AUC (%).*

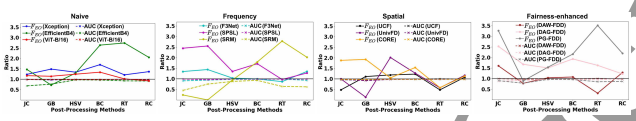

Figure 7. *Performance ratio after vs. before post-processing for all detectors on the AI-Face test set under six post-processing operations.*

Existing Benchmarks	Year	Category			Scope of Benchmark		
		Deepfake Videos	GAN	DM	Utility	Fairness	
						General	Reliability
Loc et al. [2]	2021	✓			✓	✓	
CCv1 [3]	2021	✓	✓		✓	✓	
DeepfakeBench [6]	2023	✓	✓		✓		
CDDB [10]	2023		✓		✓		
Lin et al. [8]	2024	✓	✓		✓		
Le et al. [9]	2024	✓	✓		✓		
DF40 [7]	2024	✓	✓	✓	✓		
Ours	2025	✓	✓	✓	✓	✓	✓

Table 1. *Comparison of existing AI-generated face detection benchmarks and ours. Fairness 'General' means fairness evaluation under default/basic settings. Fairness 'Reliability' measures fairness consistency across dynamic scenarios (*e.g.*, post-processing).*

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AI-Generated

Dataset	Year	Face Images		Generation Category			#Generation Methods	Source of Real Images	Demographic Annotation		
		#Real	#Fake	Deepfake Videos	GAN	DM			Skin Tone	Gender	Age
DF-1.0 [40]	2020	2.9M	14.7M	✓			1	Self-Recording		✓	
DeePhy [12]	2022	1K	50.4K	✓	✓		3	YouTube	✓	✓	✓
DF-Platter [13]	2023	392.3K	653.4K	✓	✓		3	YouTube	✓	✓	✓
GenData [25]	2023	-	20K		✓	✓	3	CelebA [41]		✓	
A-FF++ [11]	2024	29.8K	149.1K	✓			5	YouTube		✓	✓
A-DFD [11]	2024	10.8K	89.6K	✓			5	Self-Recording		✓	✓
A-DFDC [11]	2024	54.5K	52.6K	✓	✓		8	Self-Recording		✓	✓
A-Celeb-DF-v2 [11]	2024	26.3K	166.5K	✓			1	Self-Recording		✓	
A-DF-1.0 [11]	2024	870.3K	321.5K	✓			1	Self-Recording		✓	✓
AI-Face	2025	400K	1.2M	✓	✓	✓	37	FFHQ [26], IMDB-WIKI [27], real from FF++ [15], DFDC [17], DFD [28],Celeb-DF-v2 [16]	✓	✓	✓

Table 2. *Quantitative comparison of existing datasets with ours on demographically annotated AI-generated faces.*

Evaluation Type	Dataset	Gender			Age		
		ACC	Precision	Recall	ACC	Precision	Recall
Strategic	A-FF++	8.143	17.583	5.966	37.700	39.459	45.381
	AI-Face	**86.857**	**74.404**	**77.367**	**85.700**	**74.024**	**63.751**
	A-DFDC	21.600	28.604	23.082	33.400	38.011	40.165
	AI-Face	**91.700**	**92.129**	**83.448**	**77.000**	**76.184**	**62.646**
Random	A-Celeb-DF-V2	89.628	90.626	90.494	-		
	AI-Face	**91.206**	**91.474**	**91.767**	-		
	A-DFD	70.900	71.686	74.435	-		
	AI-Face	**92.300**	**91.060**	**91.727**	-		

Table 3. *Annotation quality assessment results (%) for A-FF++, A-DFDC, A-Celeb-DF-V2, A-DFD, and our AI-Face. ACC: Accuracy.*

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Measure	Attribute	Metric	Model Type											
			Naive			Frequency			Spatial			Fairness-enhanced		
			Xception [36]	EfficientB4 [37]	ViT-B/16 [38]	F3Net [18]	SPSL [19]	SRM [20]	UCF [22]	UnivFD [23]	CORE [21]	DAW-FDD [4]	DAG-FDD [4]	PG-FDD [5]
Fairness(%)↓	Skin Tone	F_{MEO}	8.836	8.300	6.264	19.938	8.055	10.002	17.325	**2.577**	10.779	14.118	6.551	6.465
		F_{DP}	9.751	**6.184**	7.728	12.876	9.379	10.897	12.581	8.556	10.317	10.706	8.617	9.746
		F_{OAE}	1.271	4.377	2.168	2.818	1.135	0.915	1.883	2.748	1.332	1.667	1.388	**0.882**
		F_{EO}	12.132	11.062	8.813	23.708	9.789	14.239	21.92	**5.536**	13.069	16.604	7.383	9.115
	Gender	F_{MEO}	3.975	5.385	5.104	4.717	4.411	6.271	5.074	4.503	5.795	5.510	5.910	**3.190**
		F_{DP}	1.691	1.725	1.344	1.864	1.827	1.957	1.736	**1.190**	2.154	2.015	2.151	1.252
		F_{OAE}	**0.975**	1.487	1.803	1.129	1.037	1.772	1.451	1.622	1.389	1.325	1.420	1.071
		F_{EO}	4.143	5.863	6.031	4.870	4.534	6.78	5.510	5.408	5.931	5.696	6.066	**3.702**
	Age	F_{MEO}	27.883	6.796	14.937	38.801	27.614	24.843	47.500	**5.436**	33.882	45.466	15.229	14.804
		F_{DP}	10.905	11.849	11.839	14.906	11.232	11.570	17.049	15.249	12.564	14.106	**9.633**	10.467
		F_{OAE}	7.265	**2.856**	6.838	10.116	7.270	6.524	11.652	3.793	8.760	11.878	5.533	5.009
		F_{EO}	42.216	**10.300**	30.795	55.032	40.943	38.528	67.545	14.148	48.729	64.384	30.182	29.585
	Intersection	F_{MEO}	10.505	17.586	9.384	21.369	10.379	15.142	20.134	**6.119**	15.34	16.565	12.178	9.578
		F_{DP}	14.511	**8.607**	11.535	17.175	13.259	15.186	17.03	14.026	14.301	14.088	11.705	14.697
		F_{OAE}	2.536	8.461	4.928	4.870	**2.464**	3.998	3.536	6.287	2.775	3.547	4.035	3.062
		F_{EO}	24.315	25.114	27.443	47.783	21.679	30.112	43.376	20.255	28.84	33.122	26.295	**18.348**
	Individual	F_{IND}	10.338	25.742	**0.022**	1.872	2.518	7.621	0.767	3.523	0.041	3.772	0.901	0.780
Utility(%)	-	AUC↑	98.583	98.611	98.69	98.714	98.747	97.936	98.082	98.192	98.579	97.811	98.771	**99.172**
		ACC↑	96.308	94.203	94.472	95.719	**96.346**	95.092	95.151	93.651	96.224	95.426	95.722	96.174
		AP↑	99.350	99.542	99.571	99.453	99.356	99.172	99.273	99.400	99.360	99.015	99.498	**99.694**
		EER↓	5.149	6.689	6.372	5.256	**4.371**	6.483	7.708	7.633	5.145	7.063	5.499	4.961
		FPR↓	12.961	20.066	16.426	14.679	13.661	15.746	13.646	18.550	13.410	16.670	14.844	**10.971**
Training Time / Epoch			**1h15min**	2h25min	2h40min	1h18min	1h20min	3h10min	5h05min	4h	1h16min	1h25min	1h17min	7h20min

Model Type	Detector	Dataset						
		CCv2 [39]		DF-Platter [13]			GenData [25]		
		Fairness(%)↓	Utility(%)↑	Fairness(%)↓		Utility(%)↑	Fairness(%)↓		Utility(%)↑
		F_{OAE}	ACC	F_{OAE}	F_{EO}	AUC	F_{OAE}	F_{EO}	AUC
Naive	Xception	1.006(+0.031)	86.465(-9.843)	6.836(+5.861)	9.789(+5.646)	81.273(-17.310)	2.539(+1.564)	13.487(+9.344)	96.971(-1.612)
	EfficientB4	4.077(+0.259)	82.980(-11.223)	8.786(+7.299)	12.370(+6.507)	67.694(-30.917)	3.304(+1.817)	**1.995** (-3.686)	93.213(-5.398)
	ViT-B/16	2.167(+0.364)	81.489(-12.983)	**0.015** (-1.788)	12.373(+6.342)	76.050(-22.640)	3.164(+1.361)	9.610(+3.579)	88.253(-10.437)
Frequency	F3Net	5.743(+4.614)	87.867(-7.852)	3.521(+2.392)	**6.445** (+1.575)	85.112(-13.602)	1.188(+0.059)	16.306(+11.436)	91.603(-7.111)
	SPSL	**0.601** (-0.436)	80.006(-16.340)	5.109(+4.072)	7.842(+3.308)	82.175(-16.572)	1.385(+0.348)	9.261(+4.272)	**98.838** (+0.091)
	SRM	7.000(+5.228)	79.768(-15.324)	3.823(+2.051)	6.567(-0.213)	66.401(-31.535)	3.281(+1.509)	7.907(+1.127)	90.049(-7.887)
Spatial	UCF	2.169(+0.718)	**93.009** (-2.142)	8.687(+7.236)	17.068(+11.558)	80.821(-17.261)	3.513(+2.062)	10.529(+5.019)	87.778(-10.304)
	UnivFD	7.625(+6.003)	67.983(-25.668)	4.540(+2.918)	9.950(+4.542)	76.443(-21.749)	1.645(+0.023)	3.848(-1.560)	94.418(-3.774)
	CORE	4.410(+3.021)	83.328(-12.896)	7.741(+6.352)	17.348(+11.417)	77.226(-21.353)	3.759(+2.370)	23.289(+17.358)	98.408(-0.171)
Fairness-enhanced	DAW-FDD	4.726(+3.401)	84.685(-10.741)	5.536(+4.211)	13.667(+7.791)	81.807(-16.004)	1.443(+0.118)	10.228(+4.532)	97.854(+0.043)
	DAG-FDD	2.364(+0.944)	83.918(-11.804)	3.064(+1.644)	22.203(+16.137)	75.206(-23.565)	**0.714** (-0.706)	10.332(+4.266)	92.108(-6.663)
	PG-FDD	1.513(+0.442)	92.852(-3.322)	4.565(+3.494)	9.717(+6.015)	**85.271** (-13.901)	3.063(+1.992)	9.479(+5.777)	93.329(-5.843)

Table 5. *Fairness generalization results based on the gender attribute. The smallest performance changes (in parentheses) and the best performance are in red and in* ***bold****, respectively. Only F_{OAE} fairness metric and ACC metric are used in CCv2 due to all samples are real.*

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